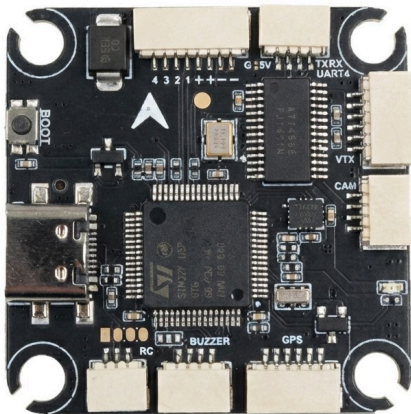
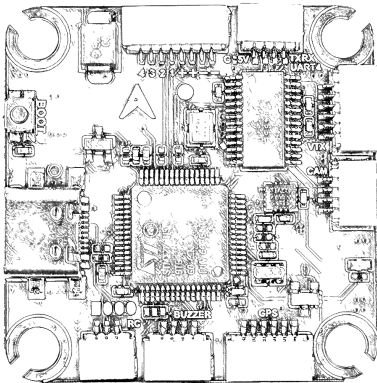


AF-F4 nano

User Guide



AF-F4 nano Overview



Compact STM32F405 flight controller with an ICM-42688-P IMU. The novaX hardware definition configures an SPL06 barometer, 8 motor/servo outputs, 1 LED output and 6 UARTs plus USB.

AF-F4 nano Specifications

• Basic Specifications

MCU	STM32F405
IMU	ICM-42688-P
Barometer	SPL06 (current novaX hardware definition)
Operating Voltage	9 – 25 V DC
Output Voltage	3.3V/1A · 5V/3A · 10V/3A
USB Input	USB Type-C (Firmware & Power)
PWM Output	8 motor/servo channels + 1 LED channel
Serial Ports	6 hardware UARTs + USB configured; verify connector pinout

AF-F4 nano Specifications

• Basic Specifications

Serial Mapping	SERIAL1 = USART1 · SERIAL2 = USART2 · SERIAL3 = USART3 · SERIAL4 = UART4 · SERIAL5 = UART5 · SERIAL6 = USART6 (USB = SERIAL0)
Servo Output	2 Port
RC Input	PWM / PPM / S.Bus
Size	39.4 × 39.4 mm
Mount Hole	30.5 × 30.5 mm / M4
Weight	9.3 g
Operating Temp	-20 ~ +70 °C
Supported F/W	novaX ArduPilot (released); Betaflight board configuration available

AF-F4 nano Pinout

• Pinout

AF-F4 nano Pinout

Both sides were straight on. Each table lists the pins left → right (side connectors: top → bottom) as you see them in the photo. The board silk screens the I2C, I2S and every bottom pad; the other top connectors keep the maker's order. There are no pin numbers.

red = power Black = GND blue = signal grey = not connected

Top side

I2C					I2S				
5-Pin: I2C 4 3 2 1 + - (V = VBAT)					5-Pin: I2S4 - I2S3 G2 G1 TX RX				
S4	S3	S2	S1	+	-	GND	S4	S3	S2



VTX	
5-Pin: UART1 TX - maker's order	
NC	
NC	
NC	
VTX	
TX	

CAM	
5-Pin: video in - maker's order	
SV	
GND	
CAM	

RC			GPS				
5-Pin: PP33 - maker's order			5-Pin: UART1 + I2C1 - maker's order				
GND	SV	PP33	SDA1	SCL1	RX1	TX1	SV GND

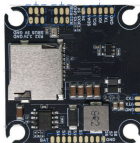
BUZZER			
5-Pin: UART1 RX - maker's order			
SV	RX3	BUZZ	GND

Bottom side (pads)

RGB pads		
UART1 / UART1 - LED or serial		
RGB	GND	SV
RX3	TX3	RX3

SERVO pads		
UART1 RX - RC input		
GND	SV	SERVO
GND	3.3V	RX2

I2C / UART pads					
I2C1 / UART1					
SV	SDA1	SCL1	RX1	TX1	GND



VTX pads	
UART1 TX - VTX control	
TX3	
VTX	
GND	

MOTOR / POWER pads							
Motor signals G2 - G8 + battery							
RED	S4	S3	S2	S1	S4	S3	GND

Also on the board: M3U STM32F405RG (168 MHz - 1 MB flash - I2S) KB (RAM) - IMU ICM-42055-P - OSD AT79SG - BOOT button and USB Type-C on the top - TF card slot and the I2C on the bottom.

AF-F4 nano Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set AHRS_ORIENTATION. Mounting: 30.5 × 30.5 mm / M4.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 9 – 25 V DC.
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-F4 nano Firmware

• Available Firmware

Type	Version	Date
ArduPilot (.apj package)	1.2.3	2026-07-11
Bootloader + App (merged HEX · DFU / SWD)	1.2.3	2026-07-11

Download files from the product page (QR).

AF-F4 nano Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.



<Detailed Manual>

